INCH-POUND

MIL-PRF-49471/7A(CR)
30 November 2000
Superseding
MIL-PRF-49471/7(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X588/U

This specification sheet is approved for use within Communications Electronics Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

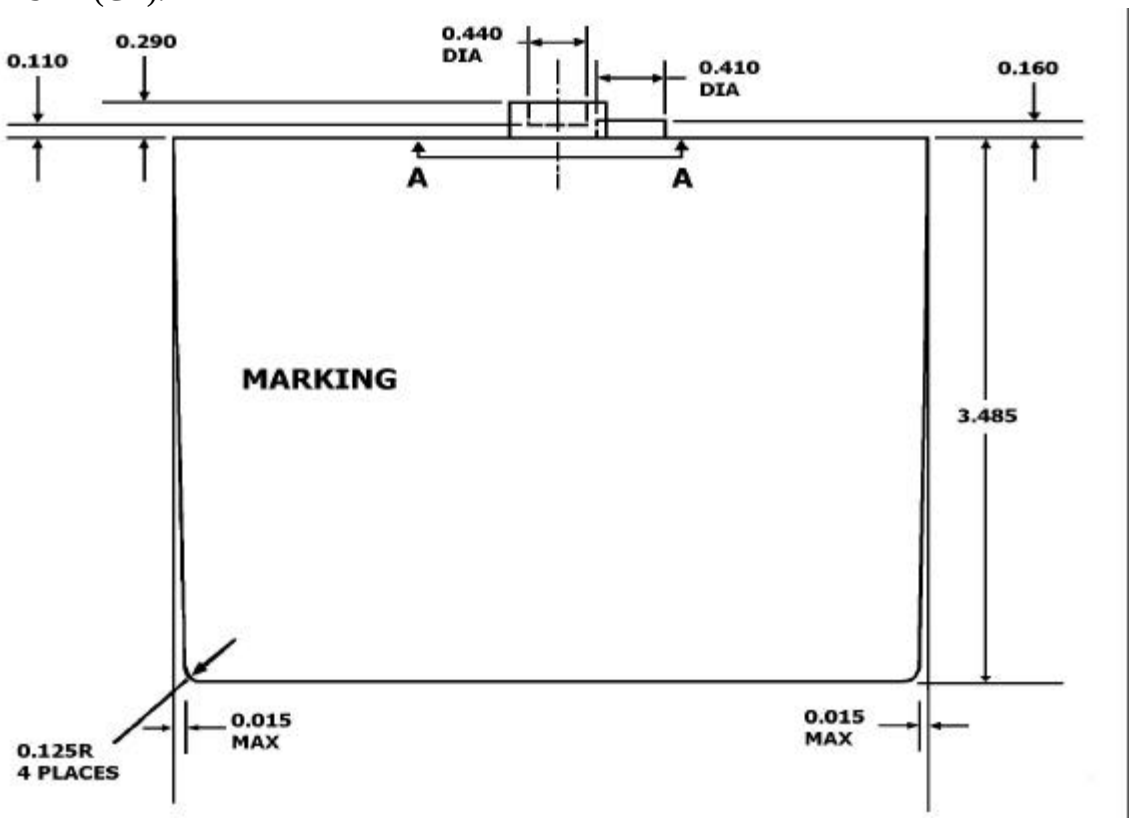


Figure 1. Battery dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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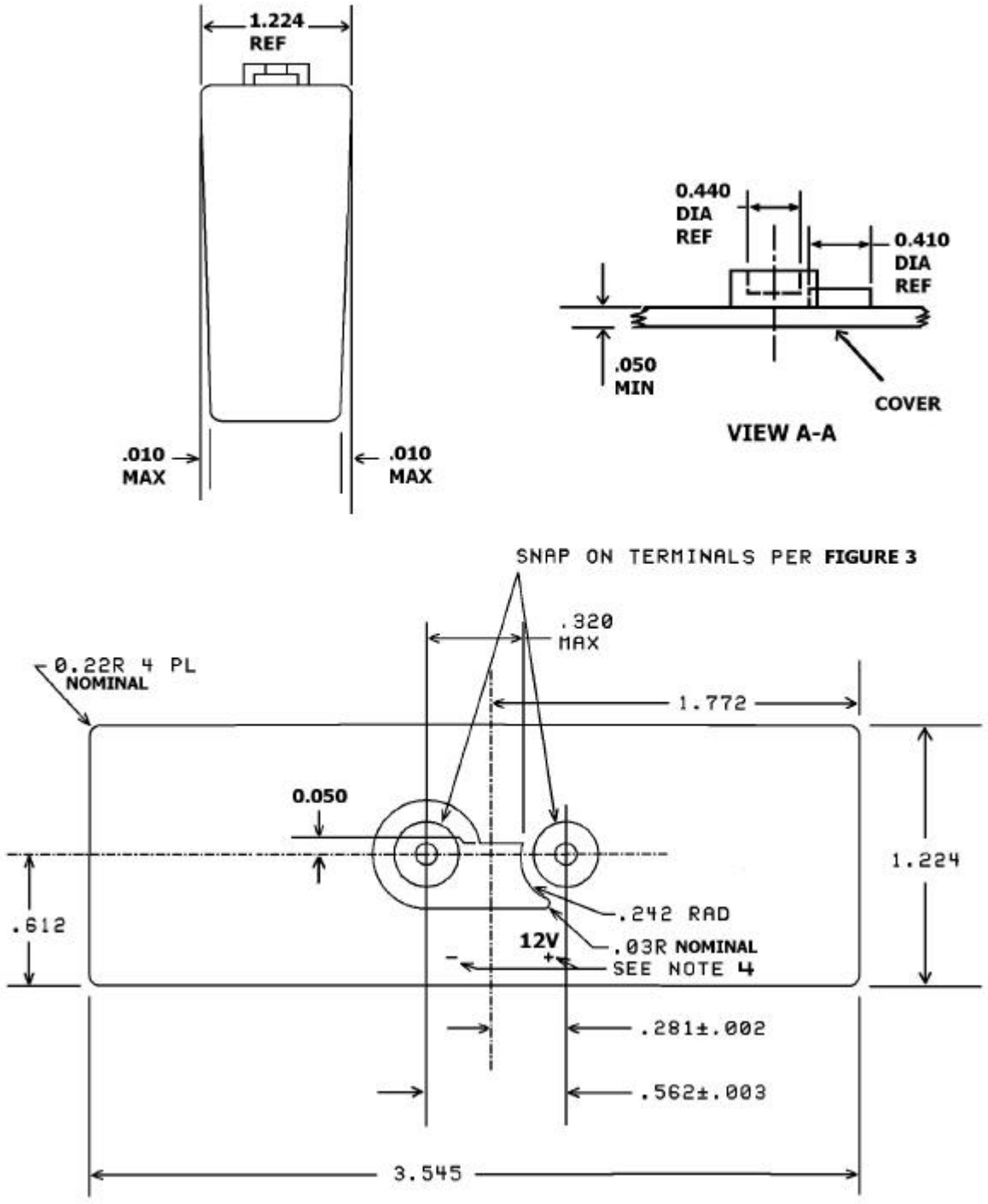
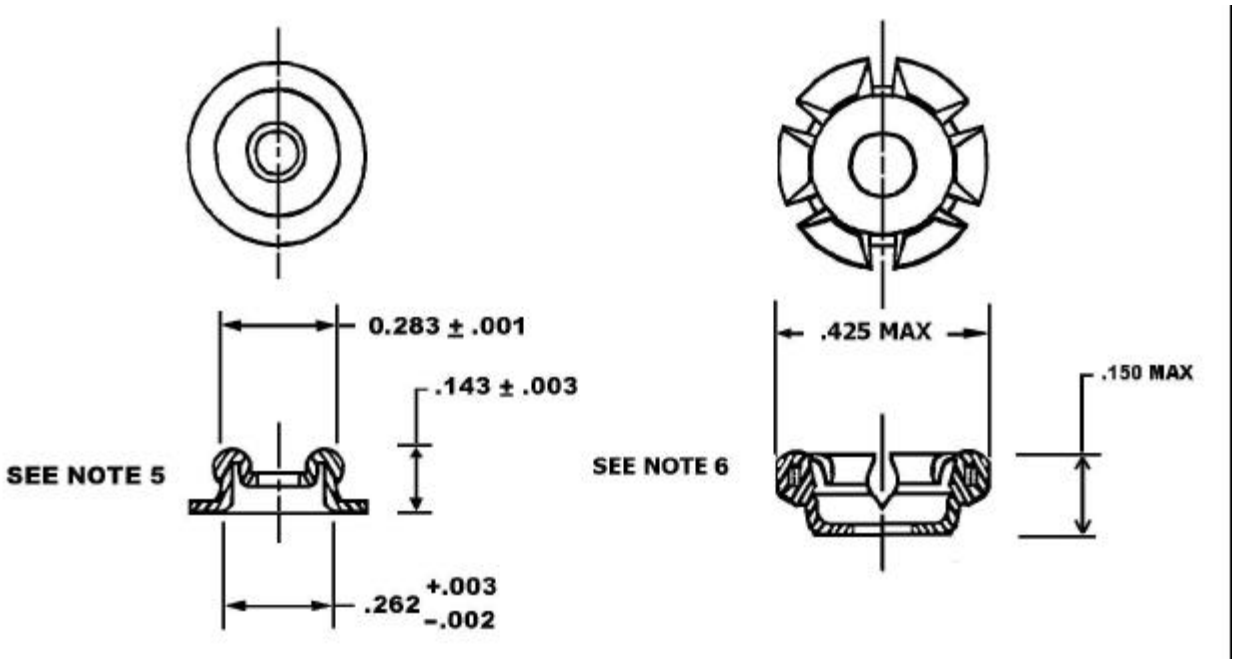


FIGURE 2. Battery Dimensions and Terminal Marking

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Stud, Positive Terminal Socket, Negative Terminal

FIGURE 3. Battery Terminals

NOTES:

1. All dimension are in inches.

2. Tolerances, unless otherwise specified, are ±.015 inch.

3. The battery connector shall be snap-on terminals per Figure 3 with protective shield (negative terminal) as shown in Figure 2.

4. Terminal and voltage marking to be embossed or impressed.

5. Stud shall be of plated soft brass not less than 0.015 inches thickness.

6. Socket shall be plated spring brass.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 & 2.

Maximum voltage: 16.5 volts.

Cut-off voltage (for capacity): 11.00 volts.

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Terminals: Snap-On type (see Figure 2).

Weight (maximum): 0.98 pounds (445 grams).

Battery short circuit test: Applicable.

Charge protection: Applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable.

State of Charge Indicator: Applicable.

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirements in Hours to 11.00 Volts
I	3.6
L	1.8
H	3.6
HT	3.1
LT	1.7
IT	3.3
LR1	53.0

Initial voltage delay: During the I, H, HT, and IT discharge tests, initial voltages below the 11.00 volt minimum voltage cannot exceed a 10 second duration. During L and LT discharge tests initial voltage below 10.00 volts cannot exceed a 20 second duration.

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METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all specified capacity tests are described in the basic specification (see 4.8.9).

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 0.8 amperes at 70±5°F to 2.0 volts. Each discharged cell shall then be connected in series to four fresh cells. Each string shall be discharged at 0.8 amperes at 70±5°F for 3.6 hours.

Batteries: For all capacity tests except the LR1, the batteries shall be discharged at 0.8 amperes constant current to cut-off volts (zero volts for the I test)..For the LR1 test, batteries shall be discharged at 65 milliamperes to 11.25 volts followed by 1.1 watts to 1.5 volts.

Forced Discharge: After discharge to zero volts on the I capacity test, the batteries shall be force discharged at 0.8 amperes for a minimum of 10 minutes.

Overload test: Batteries shall be discharged at 1.6 amperes.

Abuse test, predischarge: Designated samples shall be discharged at 0.8 amperes for 1.8 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 1.0 amperes for 1 minute followed by 0.72 amperes for 4 minutes, cycled continuously to cut-off voltage.

Closed-circuit voltage (batteries and cell strings): Load resistance of 13.75 ohms shall be used between terminals. Voltage shall be above 11.00 volts within 10 seconds.

Cell closed circuit voltage: Load resistance of 2.5 ohms shall be used. Voltage shall be above 2.2 volts within 5 seconds.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
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